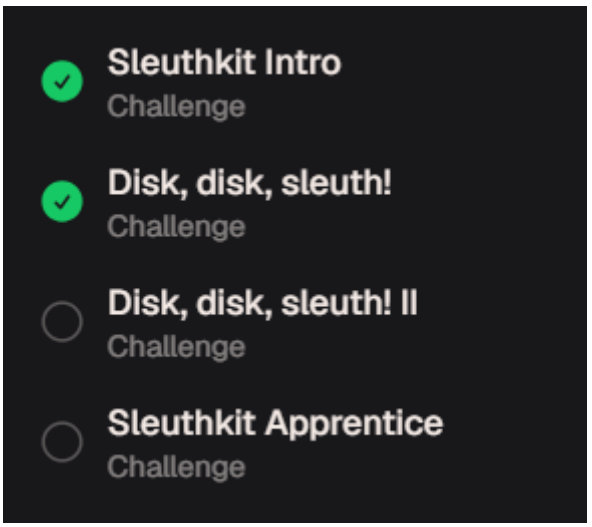
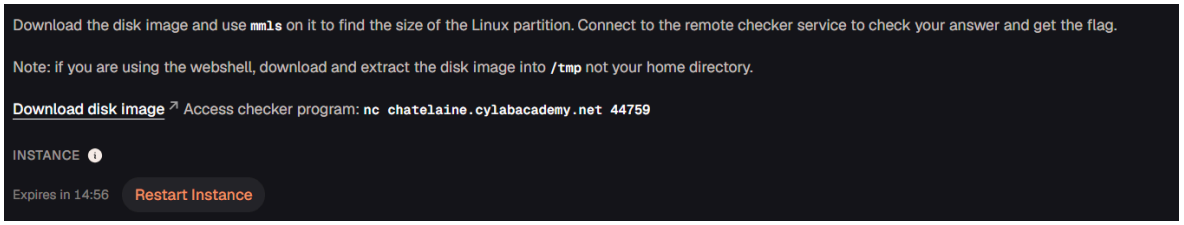


Challenge 1 : Sleuthkit Intro



File Bernama : disk.img.gz

1. Kita lihat dengan exiftool disk.img.gz karena merupakan file binary

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ exiftool disk.img.gz
ExifTool Version Number         : 13.50
File Name                       : disk.img.gz
Directory                       : .
File Size                       : 30 MB
File Modification Date/Time     : 2026:10:16 09:29:14+07:00
File Access Date/Time           : 2026:10:16 09:29:14+07:00
File Inode Change Date/Time     : 2026:10:16 09:29:31+07:00
File Permissions                : -rwxrwxrwx
File Type                       : GZIP
File Type Extension             : gz
MIME Type                       : application/x-gzip
Compression                     : Deflated
Flags                           : FileName
Modify Date                     : 2021:09:22 02:34:53+07:00
Extra Flags                     : (none)
Operating System                : Unix
Archived File Name              : disk.img
```

2. Disitu tertulis file bertipe gzip yang artinya kita perlu unzip (**gunzip disk.img.gz**) untuk mendapatkan file archived bernama **disk.img**
3. Kita gunakan mmls seperti instruksi ke **disk.img** dan fokus kepada deskripsi linux. Hubungkan ke port `nc chatelaine.cylabacademy.net 44759` dan masukkan input yang diminta sesuai deskripsi linux
4. Ketemu flagnya yakni `academy{mm15_f7w!}`

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ mmls disk.img
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  000:000   0000002048   0000204799   0000202752   Linux (0x83)
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ nc chatelaine.cylabacademy.net 44759
What is the size of the Linux partition in the given disk image?
Length in sectors: 0000202752
0000202752
Great work!
academy{mm15_f7w!}
```

Challenge 2 : Disk,Disk,Sleuth!

Use **srch_strings** from the sleuthkit and some terminal-fu to find a flag in this disk image. **dds1-alpine.flag.img.gz**

HINTS

1. Have you ever used **file** to determine what a file was?
2. Relevant terminal-fu in Challenge Library: **https://learn.cylabacademy.org/library/85**
3. Mastering this terminal-fu would enable you to find the flag in a single command: **https://learn.cylabacademy.org/library/48**
4. Using your own computer, you could use qemu to boot from this disk!

File bernama : dds1-alpine.flag.img.gz

1. Sama seperti Challenge 1, file ini merupakan gzip yang perlu kita **gunzip** untuk mencari file bernama **dds1-alpine.flag.img**.
2. Setelah menemukan **dds1-alpine.flag.img**. kita lakukan sesuai instruksi pada soal yakni `srch_strings dds1-alpine.flag.img | grep -i academy` (kita menggunakan grep -i untuk menghasilkan output search flagnya berupa academy{})

```
tzHc
^C
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$  srch_strings dds1-alpine.flag.img | grep -i academy
  SAY academy{f0r3ns1c4t0r_n30phyt3_6502313d}
```

3. Ketemu flagnya yakni `academy{f0r3ns1c4t0r_n30phyt3_6502313d}`

Challenge 3 : Disk,Disk,Sleuth! II

All we know is the file with the flag is named **down-at-the-bottom.txt**... **dds2-alpine.flag.img.gz**

HINTS

1. The sleuthkit has some great tools for this challenge as well.
2. Sleuthkit docs here are so helpful: **TSK Tool Overview**
3. This disk can also be booted with qemu!

File bernama : dds2-alpine.flag.img.gz

4. Masih sama seperti sebelumnya, kita lakukan gunzip hingga menemukan file archived yang tersedia di output **exiftool.**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ exiftool dds2-alpine.flag.img.gz
ExifTool Version Number         : 13.50
File Name                       : dds2-alpine.flag.img.gz
Directory                       : .
File Size                       : 30 MB
File Modification Date/Time     : 2026:10:16 09:52:28+07:00
File Access Date/Time           : 2026:10:16 09:52:34+07:00
File Inode Change Date/Time     : 2026:10:16 09:52:36+07:00
File Permissions                : -rwxrwxrwx
File Type                       : GZIP
File Type Extension             : gz
MIME Type                       : application/x-gzip
Compression                     : Deflated
Flags                           : FileName
Modify Date                     : 2026:09:23 09:34:12+07:00
Extra Flags                     : (none)
Operating System                : Unix
Archived File Name              : dds2-alpine.flag.img
```

5. Karena metode sleuthkit, kita coba gunakan lagi query **mmls dds2-alpine.flag.img** dan catat apapun yang menjadi deskripsi bagian **linux**
6. kita list file di partisi dengan **fls** (gunakan port **2048** pada deskripsi linux) selanjutnya kita extract file dengan **icat** (gunakan port di output sebelumnya yakni r/r **18291**)

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ mmls dds2-alpine.flag.img
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  000:000   0000002048   0000262143   0000260096   Linux (0x83)
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ fls -o 2048 -r -p dds2-alpine.flag.img | grep -i "down-at-the-bottom"
r/r 18291:      root/down-at-the-bottom.txt
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ icat -o 2048 dds2-alpine.flag.img 18291 > flag.txt
```

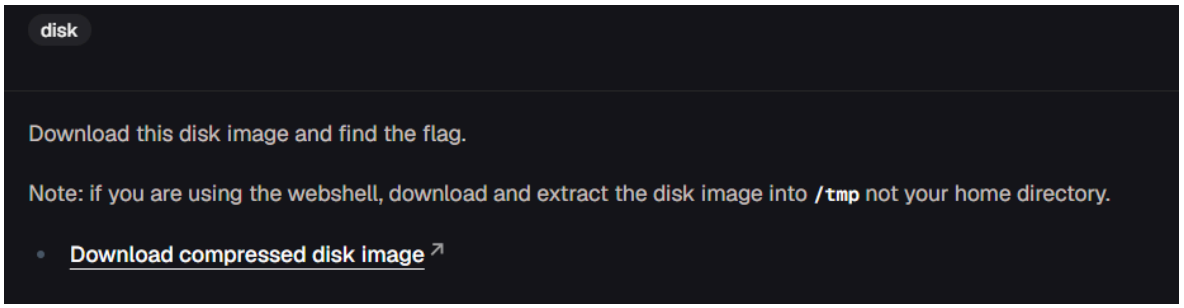
7. Tinggal kita baca file flagnya yakni **cat flag.txt** dan yap ketemu flagnya yakni `academy{f0r3ns1c4t0r_n0v1c3_95c645ad}` **(rapikan dengan query *cat flag.txt | tr -d '\n' | grep -oP '\(\K[^)]+' | tr -d '\n'*)**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ cat flag.txt
 _   _   _   _   _   _   _   _   _   _   _   _   _   _   _
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \
( a ) ( c ) ( a ) ( d ) ( e ) ( m ) ( y ) ( { ) ( f ) ( 0 ) ( r ) ( 3 ) ( n )
 \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/

 _   _   _   _   _   _   _   _   _   _   _   _   _   _   _
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \
( s ) ( 1 ) ( c ) ( 4 ) ( t ) ( 0 ) ( r ) ( _ ) ( n ) ( 0 ) ( v ) ( 1 ) ( c )
 \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/

 _   _   _   _   _   _   _   _   _   _   _   _   _
/ \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \ / \
( 3 ) ( _ ) ( 9 ) ( 5 ) ( c ) ( 6 ) ( 4 ) ( 5 ) ( a ) ( d ) ( } )
 \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/ \_/
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ cat flag.txt | tr -d '\n' | grep -oP '\(\K[^)]+' | tr -d '\n'
academy{f0r3ns1c4t0r_n0v1c3_95c645ad}lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$
```

Challenge 4 : Sleuthkit Apprentice



File bernama : disk.flag.img.gz

1. Sama kayak sebelumnya, kita lakukan gunzip dulu untuk menemukan file intinya yakni **disk.flag.img** ketika sudah lakukan metode sleuthkit **mmls disk.flag.img**

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ mmls disk.flag.img
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Primary Table (#0)
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  000:000   0000002048   0000206847   0000204800   Linux (0x83)
003:  000:001   0000206848   0000360447   0000153600   Linux Swap / Solaris x86 (0x82)
004:  000:002   0000360448   0000614399   0000253952   Linux (0x83)
```

2. Karena banyak output linux, kita fokus kepada **linux (0x83)** dan kita lakukan list file di **partisi pertama** dengan port **2048**.
3. Karena tidak menghasilkan apa apa pada partisi pertama, kita lanjut list file di **partisi kedua** dengan port **360448**.

```
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ fls -o 2048 -r -p disk.flag.img | grep -i academy
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ fls -o 2048 -r -p disk.flag.img
d/d 11: lost+found
r/r 12: ldlinux.sys
r/r 13: ldlinux.c32
r/r 15: config-virt
r/r 16: vmlinuz-virt
r/r 17: initramfs-virt
l/l 18: boot
r/r 20: libutil.c32
r/r 19: extlinux.conf
r/r 21: libcom32.c32
r/r 22: mboot.c32
r/r 23: menu.c32
r/r 14: System.map-virt
r/r 24: vesamenu.c32
V/V 25585:      $OrphanFiles
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ fls -o 360448 -r -p disk.flag.img | grep -i flag
r/r * 2082(realloc):    root/my_folder/flag.txt
r/r 2371:       root/my_folder/flag.uni.txt
lenovo@LAPTOP-8KDUGB9T:/mnt/c/Users/LENOVO/Downloads/ResumeCyber/forensicCTF-2$ icat -o 360448 disk.flag.img 2371 > academy.txt
```

4. Dan yap ketemu folder flagnya, lalu kita ekstrak foldernya dengan metode **icat** ke file **academy.text**. tinggal kita baca dengan query **cat academy.text** dan ketemu flag : `academy{by73_5urf3r_85e9b307}`